

LSTM - Training Results

Summary

An LSTM was trained to classify text into 5 toxic categories using toxic_data.csv, with the same data preparation as the RNN: each input is the query joined with its image description, duplicate rows were removed before splitting the data into 70% train, 15% validation and 15% test, and 4 classes with only 2-5 unique examples were excluded. The model uses 64 LSTM units, dropout 0.3, class weights and early stopping. On the test set it reached a weighted F1 of **0.927**, meeting the F1 ≥ 0.85 requirement, and a macro F1 of **0.814**, which is below it. The gap comes from one class: Unknown S-Type (13 test rows) shares its image descriptions with Safe, so the model often confuses the two. The other four classes score 0.92-1.00.

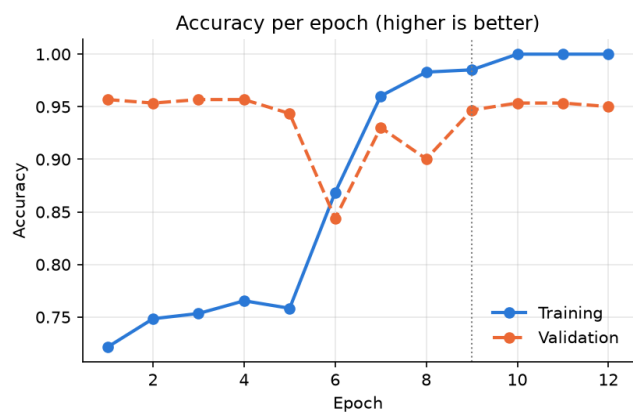
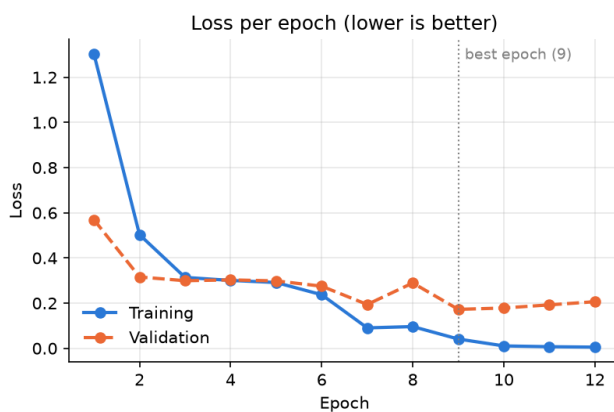
Test results

	Macro	Weighted
F1	0.814	0.927
Precision	0.814	0.927
Recall	0.814	0.927

Class	Precision	Recall	F1	Test rows
Non-Violent Crimes	1.000	1.000	1.000	31
Safe	0.924	0.917	0.920	132
Unknown S-Type	0.154	0.154	0.154	13
Violent Crimes	0.990	1.000	0.995	104
unsafe	1.000	1.000	1.000	22

Training and validation curves

LSTM training history



Confusion matrix (test set)

